

or getting disoriented in the shuffle of busy pedestrians. That is where a personal navigation device, which is a very small GPS locator, plays an important role.

As the user walks down the street, the device speaks directions and locations, so the user always knows where he or she is and where he or she is headed. In addition, the user can plan and store routes and tag locations for later reference.

Designed as an affordable GPS accessory (and not a replacement) to the cane or guide dog, the device offers an incredible amount of

gal, safe and affordable car, the sensors and innovations coming out of the project are sure to help in other areas. Not only could this serve as a means of independent, safe transportation for the visually impaired, but also for many individuals with physical and mental handicaps that prevent them from operating a motor vehicle. Whether it is further study of brain re-mapping and alternate sensory input, practical implications of better sensors and safety devices that can be applied to all motor vehicles or an all-out safe and practical means of driving blind, this technology excites many.

A group of researchers have developed a highly-sophisticated, highly-functional prosthetic arm and the results are amazing—a less-than-3.6kg prosthetic arm with such precision and control that it can peel a grape. The arm supports a number of customisable controls and modular components, making it easy to tailor to the wearer's individual needs, whether he or she requires only a hand or an entire arm and shoulder socket.

Another promising feature of the arm is its sensory feedback system. Hand sensors measure the strength of the hand grip, for instance, and provide feedback to the wearer in the form of vibrations that grow stronger as the strength of the grip increases. This enables the wearer to learn what level of grip strength is appropriate for specific tasks and how much pressure is being applied.

Development on the cochlear implant first began in the late 1950s, and the first commercial implant device received approval in 1984. However, cochlear implants have come a long way. Initially, the single-channel implant provided mostly static, while early commercial implants with five channels allowed for some indication of cadence and rhythm.

Today's cochlear implants, however, have more than 20 sound

channels, allowing wearers to hear with much better quality. The implants are still far from perfect, with background noise continually being a problem, but the technology has advanced to such a point now that voices can be heard with enough clarity to be readily understood and identified, making verbal communication possible and productive.

Cochlear implants are pretty amazing and only grow more so as the software and hardware continue to improve. More than just a hearing aid, these first pick up a sound via a microphone, which carries the signal to a small computer worn behind the ear, where it is transferred to a digital signal and transmitted to the implant itself. Once received by the implant, the device directly stimulates the auditory nerve, providing an entirely new means of auditory sensory input.

Stairs are nearly everywhere and navigating these in a traditional wheelchair is impossible. Enlisting the services of others to drag or carry the person up and down the stairs is dangerous, inconvenient and often embarrassing. iBot is a self-balancing, stair-climbing wheelchair for the physically disabled. Using self-balancing technology, it aims to give wheelchair-bound individuals the freedom to navigate any terrain.

Advancements in wearable technology enable people with disabilities to carry on with everyday functions. Winkymote is an infrared remote control hidden in make-up that helps quadriplegics change TV channels by blinking. This development has grown to include metalised false eyelashes that can activate various electronic devices and other cosmetics tech products.

People with epilepsy experience seizures at any time, often without warning. In May 2013, senior engineering students at Rice University in Texas developed Seizure Monitoring and Response



The DEKA robotic arm (Image courtesy: www.dekaresearch.com)

security, confidence and a wealth of useful information, allowing blind people to travel independently without the fear of getting lost or wandering in the wrong direction.

Speaking of mobility for the blind, engineers are developing a car that can actually be driven by the blind. The aim is to integrate several computer systems, sensors and cameras to observe the environment around the vehicle and provide alternate forms of sensory input, including sound and vibration. This may include seat vibrations of various strengths and locations, pulsing vibration signals in gloves worn by the driver, auditory alerts from a headset and a sort of screen that paints a virtual picture of the surroundings using compressed air.

The technology is extremely promising, and even if the vehicle never makes it to the street as a le-